

GATHERING OTHER- REGARDING PREFERENCES USING SOCIAL MEDIA BEHAVIOUR

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1 ABSTRACT

This paper aims to test the feasibility of using social media behaviour to gather other-regarding preferences as an alternative to the standard Berg et al. (1995) trust games as well as testing for behavioural differences between Facebook, Instagram and Twitter users. Finally, the paper aims to investigate the relationship between age, prosocial behaviour and social media behaviour. The study makes use of a standard Berg et al. (1995) trust game and then compares the findings to survey responses, which contained questions relating to other-regarding preferences and common choices found in a social media environment. The results show that there are behavioural differences between the different social media platform users but it is not possible to reliably predict trust and reciprocity preferences using social media behaviour with the approach that was used in this study. It could, however, be possible that aspects of social media, other than the ones used in this study, could be used to predict behaviour outside of social media.

2 INTRODUCTION

Trust and reciprocity as other-regarding preferences play an important role in economic theory as well as many economic models. Zak & Knack (2001) shows that interpersonal trust has a drastic impact on economic growth and that a certain degree of interpersonal trust is essential for economic development to take place. It is, therefore, important that policymakers are capable of establishing current interpersonal trust levels reliably, quickly and cheaply.

The foundation for research on trust and reciprocity as economic concepts was established by Berg, Dickhaut, & McCabe (1995). In their findings, they concluded that self-interest alone cannot be used to explain economic behaviour and that other-regarding preferences played an integral role in decision making between individuals. Social history and prospects of future interaction were also found to be key determinants of economic behaviour by Leider, Mobius, Rosenblat, & Do (2008). Social history is formed when one individual has dealt with another in the past and has formed experiences based on these past dealings. Prospects of future interactions are when one individual will likely have to deal with another individual again in the future.

Currently, the most widely-used experimental framework for gathering these other-regarding preferences is the Berg et al. (1995) trust game, which makes use of real money to ensure the experiment would be incentive compatible.

These games have two serious disadvantages; firstly, it is time-consuming since the experiment needs to be run by the experimenter in a controlled environment with all necessary preparations that go along with that, and secondly, it is expensive to run, since the experimenter will need to give payouts to (at least a number of random) participants to ensure that the experiment is incentive compatible.

This paper is about testing the feasibility of using social media behaviour to gather other-regarding preferences as an alternative to the standard Berg et al. (1995) trust games, to investigate trust and reciprocity behavioural differences between the different social media platforms and to test the relationship between age and prosocial behaviour. Using social media behaviour as an alternative to gather other-regarding preferences would be less time-consuming since it would only be necessary to observe the participants' social media behaviour, and less expensive to run since payouts to random participants would not be necessary. Because of these benefits, it would allow for analysis of large samples of data, across countries and contexts.

3 LITERATURE REVIEW

3.1 TRUST & RECIPROCITY

Trust and Reciprocity are fundamentals of prosocial behaviour with prosocial behaviour being characterised by the exchange of favours between non-related individuals (Thaler, 2016).

According to Fehr & Gächter (1998), trust and reciprocity also act as powerful effort elicitation devices, giving individuals a strong motivational drive, and can be the explanation behind various theories contradicting empirical findings. These prosocial behaviours provide justification for actions that are not based on self-interest. Reciprocity was further identified as one of the essential forces that sustains social customs.

3.2 OTHER-REGARDING PREFERENCES

Leider, Mobius, Rosenblat, & Do (2008) conducted a large online experiment on social networks to segment prosocial giving into three different categories: baseline altruism towards randomly selected strangers, directed altruism that favours friends over strangers and giving, motivated by the prospect of future interaction. The study was conducted in two parts. The first part made use of a modified dictator game with exchange rates that varied while the second part consisted of a new helping game. Decision-makers first decided on allocations between themselves and other unnamed participants. After a few days, the decision-makers again had to make an allocation but this time between themselves and named participants. Both of these allocations were repeated at varying social distances. A randomly selected decision, with its corresponding allocation, was selected each time for payment, with all participants being informed of the payment by email.

The study found that directed altruism increased by 52% when giving to friends, relative to random strangers, giving increased by 24% when the prospect of future interaction was present and giving increased when it was socially optimal.

3.3 TRUST GAMES

Berg, Dickhaut, & McCabe (1995) designed a game replicating the real-life aspect of trust and reciprocity where trust can lead to mutual gains, finding evidence of both trust (sending money to a partner) and reciprocity (returning money to someone who has sent money). The procedure (explained in more detail later) in which this was tested, consisted of participants in two rooms, room A and room B, where room A participants had the choice of sending a portion of money (if any) to participants in room B. Participants in room B would then receive triple the sent amount and then had the choice of returning a portion (if any) to the participants in room A. If participants of the trust game had strictly increasing, indirect utility functions of wealth, the dominant strategy for participants in room B would be to keep all the money instead of returning a portion to participants room A. Likewise, if participants in room A were to infer the dominant strategy of participants in room B, to send nothing in the first place, would again be the sub-game perfect equilibrium.

The deviation away from the sub-game perfect equilibrium (to send/return nothing) is what makes the positive transfers meaningful, since this behaviour deviates from the self-interested equilibrium, allowing for the interpretations of trust and reciprocity. These findings formed the foundation for research on trust and reciprocity as economic concepts and allowed these authors to conclude that self-interest alone cannot be used to explain economic behaviour. A double-blind procedure, which guaranteed absolute anonymity for the subjects and the experimenter; and a one-shot procedure, where each subject had only one chance to make a decision, were implemented as controls in the experiment. These controls ruled out contractual precommitments or reputation as explanations for the subject's behaviour. The social history treatment (treatment where subjects had some prior form of interaction with each other) that was performed showed that some subjects reciprocated due to being trusted. It can be concluded that social history plays a role in the subjects' trust and the likelihood for subjects to reciprocate.

Johnson & Mislin (2011) conducted a meta-analysis of 162 replications of the Berg et al. (1995) trust game. The meta-analysis ranged across countries, subject pools and contexts with the study consistently finding evidence of trust and reciprocity in decisions in these games. The study found that even minor variation in the standard trust game protocol can cause significant changes in measured trust behaviour. Participants were found to send less money if the payments were made at random due to the randomness of payment being treated as an added risk to the game.

Participants also returned less if they were students, who represent a younger subset of the average population, presumably due to the fact that younger people give less and are less generous than older people (see “Effect of Age on Trust and Reciprocity” below). A tendency to send less on average was observed if the participants played the role of both the sender and the receiver in the same game due to feeling less responsible, causing more selfish behaviour.

3.4 EFFECT OF AGE ON TRUST AND RECIPROCITY

In a study that made use of the standard trust game, conducted by Sutter & Kocher (2007), it was found that there was an increase in trust and reciprocity dependant on the participants' age. Six age groups were tested, ranging from 8 – 68 years in age, with older participants both offering and returning more money in the trust game. There is thus an increase in prosocial behaviour with age, stabilising around 16 – 22 years of age.

Another study conducted by Van Den Bos, Westenberg, Van Dijk & Crone (2010) confirmed this increase in prosocial behaviour with age. The development of two types of prosocial behaviour were investigated (trust and reciprocity) with four age groups ranging from 9 – 25 years in age. The study found that there were age-related changes in sensitivity to trust and reciprocity. Children earned more as the second player of the trust game (due to not reciprocating as often) while adults earned less (due to reciprocating more often). This can be explained as a decrease in ‘rational self-interest’ with age, potentially due to an increase in showing socially desirable behaviour.

4 METHODOLOGY AND DATA

4.1 GENERAL:

Participants in the study were required to complete a survey and a standard Berg, Dickhaut & McCabe (1995) trust game. Trust and reciprocity preferences were gathered through the trust game and were then compared to the survey responses to see if there is a relationship. Both the survey and the trust game were completed online through google forms (link to google forms experiment can be found in the appendix). Participation in the study was completely voluntary with all participants agreeing to the provided informed consent. Anonymity of participants were also guaranteed. The study was therefore in line with all ethics requirements of the University of Pretoria.

4.2 SAMPLING:

A sample of 266 students from the University of Pretoria were recruited to participate in the study (197 first years and 69 third years), with 223 observations remaining after removing irregularities, invalid responses and minors.

Observations where instructions weren't followed correctly, were removed, for example, participant sent more money than what the question allowed or values were not sent/returned in multiples of 10. Observation results of the dependent variables ("Trust" and "Reciprocity") with values greater than the question allowed were also discarded.

4.3 DATA GATHERING:

4.3.1 Survey

The survey contained questions relating to other-regarding preferences and common choices found in a social media environment. The questions were divided into three sections; Facebook, Instagram and Twitter. Participants were only required to answer one of the three sections based on their most commonly used social medium. Questions related to the frequencies of the different social media behaviours ranged from "Always" to "Never" with "Indifferent" being added where relevant (all survey questions can be found in the appendix).

4.3.2 Trust Game

A standard Berg et al. (1995) trust game was also played where each participant played the role of both 'sender' as well as 'receiver'. To facilitate this each participant had a sender and receiver number allocated to them, effectively allowing participants to play two games, once as a 'sender' and once as a 'receiver'. The 'sender' in this game was endowed with R100 and had to decide what portion (if any) of this money to transfer to another player, the 'receiver'. The 'receiver' then receives triple the sent amount, and can then decide how much of this tripled amount (if any) to return to the 'sender'. Sent amounts are usually interpreted as indicating trust while returned amounts are usually interpreted as indicating reciprocity. All the participants of the game were allocated a random 'sender' and 'receiver' number. At the end of the game, five random 'sender'/'receiver' numbers were drawn and the players with these numbers were paid in real money (based on the decisions they made in the trust game). This improved incentive compatibility by encouraging players to make all decisions as if they were playing for real money.

The first question in the trust game was about what portion of R100 the participant, acting as the 'sender', would send to the other participant with the corresponding 'receiver' number. Participants were given instruction to only send multiples of 10 up to a maximum of the full R100. The responses of this question converted into a percentage, made up the dependent variable used for testing trust behaviour in the model. The subsequent questions were about the participants' return responses, given each possible sent amount from R10-R100 multiplied by 3. For example; if the 'sender' sent R10, R30 would be received by the 'receiver' who can then decide what portion of the R30 be returned to the 'sender'. The average of the responses to these questions converted into a percentage made up the dependent variable used for testing reciprocity behaviour in the model.

4.4 MODELLING:

Tobit modelling was used to analyse the results of the survey and the trust game. Tobit modelling is typical in the literature on trust games due to the truncated nature of response data in these games. The model was censored with a lower limit of 0 and an upper limit of 1 since all sent and returned values (percentages) fell within this interval. The dependent variables consisted of sent/returned amounts converted into a percentage of the total amount the participant could send/return in each case (taking on a value between 0 and 1). A dummy approach was followed with the survey data so that the dummy variable took a value of 1 where respondents select the upper two options for each question (i.e. “Always” and “Usually” equals 1, 0 otherwise). This approach was used due to the tendency of South Africans to choose higher options when participating in surveys. This approach also ensured that there was enough variation in the dummies.

4.4.1 Controls

Gender, perceived relative income, ethnicity and age were included as controls, with particular interest in age. Female was used as the base for the gender dummy (equal to 1, 0 otherwise). The dummy variable for participants', perceived average income was above-average income acting as the base (IncomeAbove), with average and below-average income being grouped together. “Black” was used as the base for ethnicity (equal to 1, 0 otherwise). Participants below the age of 18 were dropped to avoid testing on minors and AgeOlder was used as the dummy variable for age, participants older than 20 years in age were used as the dummy base (equal to 1, 0 otherwise). It is expected that there will be a positive relationship with age between both Trust and Reciprocity, as found by Sutter & Kocher (2007).

4.4.2 Trust Variables

As noted previously, the dependent variable for testing whether trust, as measured in the trust game, can be predicted from trusting social media behaviour. “Trust” is the amount the participants sent from question 1 of the standard trust game (“How much of your R100 do you want to send to the player whose number is on the paper?”) converted into a percentage of the total amount the participant could send (i.e. sent amount/100). Higher sent amounts indicates higher levels of trust while lower sent amounts indicate lower levels of trust (Berg, Dickhaut & McCabe, 1995).

Four questions from each survey section (Facebook, Instagram and Twitter) were used in setting up the trust explanatory variables. The first variable, named “*CloselyLike*”, contains the dummy results from the question “Will you like someone’s post if you know them closely?” from each section. This was identified as the standard question from each section most likely to indicate trust while the second variable, named “*NotCloselyLike*”, contained the counterpart dummy responses of the first variable (i.e. “Will you like someone’s post if you don’t know them closely?”).

The third variable, named “*CloselyComment*”, contains the dummy responses to the question “Will you comment on someone’s post(s) if you know them closely?” which was expected to signify a higher level of trust due to the added time and effort the participant would invest by commenting on the post instead of just liking the post. The fourth variable, named “*NotCloselyComment*”, is the counterpart to variable 3 (“*CloselyComment*”) and contains the question “Will you comment on someone’s post(s) if you don’t know them closely?”.

4.4.3 Hypotheses:

It might be expected that commenting acts as a higher level of trust, since it would take far more of the user’s time to comment on someone’s post than to just like the post, potentially indicating a greater level of dedication to the person who made the post on the part of the commenter. The comment variables are therefore expected to indicate a higher level of trust than the like variables. Older social media users are also expected to trust more often, thereby having a larger likelihood to like and comment more often, than younger users (Sutter & Kocher, 2007 and Van Den Bos, Westenberg, Van Dijk & Crone, 2010).

The model specification for testing trust is therefore:

$$\text{Trust} = \beta_0 + \beta_1 \text{CloselyLike} + \beta_2 \text{NotCloselyLike} + \beta_3 \text{CloselyComment} + \beta_4 \text{NotCloselyComment} + \beta_5 \text{Female} + \beta_6 \text{IncomeAbove} + \beta_7 \text{EthnicityB} + \beta_8 \text{AgeOlder} + \mu$$

4.4.4 Reciprocity Variables

The dependent variable for testing reciprocity, named “*Reciprocity*”, consisted of the amounts the participants returned from questions 2-11 of the standard trust game (“If the sender sent R10, then I received R30 and I will return R__”) converted into a percentage of the total amount the participant could return in each question (i.e. returned amount/30) and then the overall average of all the questions were taken, giving the average percent of the transfer that is returned to the sender. Higher returned amounts indicate higher levels of reciprocity while lower returned amounts indicate lower levels of reciprocity (Berg, Dickhaut & McCabe, 1995).

Three questions from each survey section (Facebook, Instagram and Twitter) were used in setting up the reciprocity explanatory variables. The first variable, named “*LikeRecent*”, contains the dummy results from the question “If someone has liked your most recent post, will you like their recent post(s)?” from each section. This was identified as the standard question from each section most likely to indicate reciprocity while the second variable, named “*NotLikeRecent*”, contained the counterpart dummy responses of the first variable (i.e. “If someone has not liked your most recent post, will you like their recent post(s)?”).

The third variable, named “*AlwaysLiked*”, contains the dummy responses to the question “Will you like someone’s recent post(s) if they have always liked your posts?”

4.4.5 Hypotheses:

The *AlwaysLiked* variable might be expected to have a stronger relationship since if someone has always liked your posts, there is a strong prevalence of social history which according to Berg et al. (1995), should have a strong positive effect on prosocial behaviour. It is also expected that older people would like posts more often due to the increase in prosocial behaviour as age increases (Sutter & Kocher, 2007 and Van Den Bos, Westenberg, Van Dijk & Crone, 2010).

The model specification for testing reciprocity is therefore:

$$\text{Reciprocity} = \beta_0 + \beta_1 \text{LikeRecent} + \beta_2 \text{NotLikeRecent} + \beta_3 \text{AlwaysLiked} + \beta_4 \text{Female} + \beta_5 \text{IncomeAbove} + \beta_6 \text{EthnicityB} + \beta_7 \text{AgeOlder} + \mu$$

4.5 TREATING VARIABLES AS CONTINUOUS:

The explanatory variables were also converted into continuous variables and regressed with normal OLS for the purpose of a robustness check.

Table 1: Dummy Variable Summary

Dummy Variable	Question
CloselyLike	“Will you like someone’s post if you know them closely?”
NotCloselyLike	“Will you like someone’s post if you don’t know them closely?”
CloselyComment	“Will you comment on someone’s post(s) if you know them closely?”
NotCloselyComment	“Will you comment on someone’s post(s) if you don’t know them closely?”
LikeRecent	“If someone has liked your most recent post, will you like their recent post(s)?”
NotLikeRecent	“If someone has not liked your most recent post, will you like their recent post(s)?”
AlwaysLiked	“Will you like someone’s recent post(s) if they have always liked your posts?”

5 EMPIRICAL RESULTS

5.1 GENERAL

From Figure 1 below it is clear that there was a greater preference for Instagram, making up 57.52% of all participants and that Twitter was the least preferred platform, making up only 13.53% of all participants with Facebook in the middle, making up 21.43% of all participants.

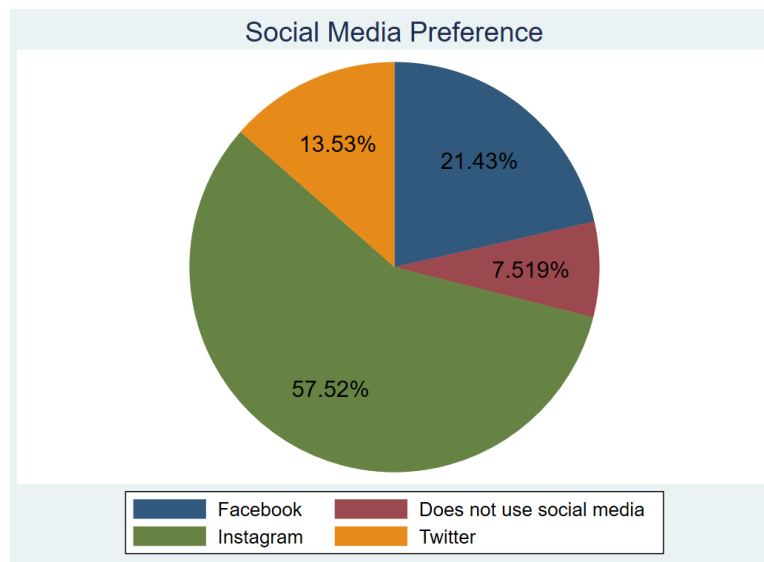


Figure 1: Pie Chart of Social Media Preference

The mean percent transfer of the 'Trust' variable for the combined model was 49.94% (0.4994196) and the mean percent return of the 'Reciprocity' variable for the combined model was 43.10% (0.4310674). The meta-analysis conducted by Johnson & Mislin (2011), where 775 South Africans' mean percent transfer and mean percent return was tested found that the mean percent transferred for South Africans was 0.44 (44%) and the mean percent returned was 0.24 (24%). Compared to the meta-analysis, the trust variable is slightly higher than usually found in South Africa, while the reciprocity variable is much higher than usual. Using a student sample with the strategy method, where participants responded to multiple possible sent amounts, might explain these differences.

Table 2: Mean Percent Transferred & Returned

	Combined	Facebook	Instagram	Twitter
Mean Percent Transferred	0.4994196	0.4734694	0.4989516	0.48125
Mean Percent Returned	0.4310674	0.4587304	0.4269382	0.406634

Table 2 shows that the mean percent transfers were roughly the same across all social media platforms, with it being slightly higher with the participants that chose Instagram and slightly lower with the participants that chose Facebook.

Mean percent returns were also roughly the same across all platforms with it being slightly higher with participants that chose Facebook and slightly lower with participants that chose Twitter.

A Wilcoxon rank-sum test was run on the mean percent transfers and returns but no differences were found to be significant even at 10% level of significance.

Table 3: Mean Values for Dummy Variables

	Combined	Facebook (A)	Instagram (B)	Twitter (C)
CloselyLike	.7053571 _{AC}	.6734694	.8629032	.5625
NotCloselyLike	.2142857	.1836735	.2419355	.28125
CloselyComment	.3303571	.3061224	.3629032	.4375
NotCloselyComment	.0714286 _{BC}	.0816327	.0483871	.1875
LikeRecent	.2142857 _{AC}	.3265306	.2258065	.125
NotLikeRecent	.2142857 _{AB}	.122449	.2822581	.21875
AlwaysLiked	.4598214	.5306122	.516129	.40625

*The most significant differences between the mean values of the social media platforms, at a 5% l.o.s. after running a Wilcoxon rank-sum test, are indicated by the relevant subscripts - A, B and C.

From Table 3 it is evident that Facebook users are far more likely than Twitter users to always or usually like someone's post if they know them closely (67.35% vs 56.25% of the time). This could possibly indicate that Facebook users value trust more than Twitter users. Twitter users were found to be far more likely than Instagram users to always or usually comment on someone's posts if they don't know that person closely (18.75% vs 4.84% of the time) which is most likely due to the nature of Twitter; where more emphasis is placed on commenting on shared ideas.

Facebook users were also found to be far more likely than Twitter users to always or usually like someone's post if that person has liked their most recent post (32.65% vs 12.5% of the time), possibly indicating that Facebook users value social history and reciprocity more than Twitter users. On the other hand, Instagram users were found to be far more likely than Facebook users to always or usually like someone's post if that person has not liked their most recent posts (28.23% vs 12.24% of the time).

Taking the above mentioned into account, it is clear that there are behavioural differences between the different social media platform users and that there are possible differences in likelihood to trust/reciprocate in the social media setting, depending on what social media platform is used.

5.2 TRUST:

Table 4 shows the coefficients and standard errors of each of the explanatory variables used in the different regressions as well as their significance level.

Table 4: Tobit Regression Results for Testing Trust with 223 Observations

	Combined	Facebook	Instagram	Twitter
Trust				
CloselyLike	-0.048 (0.052)	-0.050 (0.108)	0.071 (0.084)	0.164 (0.119)
NotCloselyLike	-0.046 (0.055)	0.206 (0.127)	-0.097 (0.065)	0.017 (0.139)
CloselyComment	-0.047 (0.051)	-0.131 (0.105)	-0.036 (0.061)	-0.178 (0.156)
NotCloselyComment	0.091 (0.089)	-0.324* (0.176)	0.287** (0.139)	0.096 (0.180)
Female	0.028 (0.046)	-0.027 (0.104)	0.031 (0.057)	0.116 (0.130)
IncomeAbove	0.042 (0.050)	-0.092 (0.115)	0.062 (0.058)	0.647*** (0.212)
EthnicityB	-0.118** (0.048)	-0.190* (0.106)	-0.161** (0.066)	0.036 (0.160)
AgeOlder	0.011 (0.047)	-0.003 (0.102)	-0.042 (0.065)	0.171 (0.126)
Constant	0.588*** (0.056)	0.709*** (0.114)	0.490*** (0.086)	0.210 (0.162)
Observations	223	49	124	31
LR chi2	10.60	10.84	14.25	16.40
Prob > chi2	0.2256	0.2111	0.0754*	0.0370**

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

NotCloselyComment was found to be statistically significant at a 10% level of significance (l.o.s.) for both Facebook and Instagram and have coefficients of -0.324 and 0.287 respectively. This shows that commenting on someone's post that you don't know closely has a small negative predictive effect on trust when looking at Facebook but a small positive predictive effect when looking at Instagram. In the social media survey, no distinction between positive and negative comments were made. It is expected that there would be a positive relationship between positive comments and trust and a negative relationship between negative comments and trust.

A possible explanation for the *NotCloselyComment* variable results could, therefore, be that Instagram asks the users whether they are sure they want to post the comment before posting it (acting as an indirect suggestion), in an attempt to discourage users from sending malicious comments to each other. This ‘nudge’ to only post positive comments could explain why Instagram users have a small positive relationship between trust and commenting on someone’s post they don’t know closely. With Facebook, however, there is no indirect suggestion to discourage malicious comments. This could then be the cause of the small negative relationship between commenting on someone’s post that you don’t know closely and trust.

The income control variable, *IncomeAbove*, was found to be statistically significant at a 1% l.o.s. when looking at Twitter and had a coefficient of 0.647 which shows that there is a moderate positive relationship between perceived above-average income and trust under Twitter users. Twitter users who perceived themselves as having above-average income will, therefore, transfer more on average in trust games. This indicates higher levels of trust as established by Berg et al. (1995).

The ethnicity control variable, *EthnicityB*, was also significant at a 1% l.o.s. for the combined model, significant at a 10% l.o.s. when looking at Facebook and significant at a 5% l.o.s. when looking at Instagram with a coefficient of -0.118, -0.190 and -0.161 respectively. *EthnicityB* has an, albeit small, negative relationship with trust. This means that black participants will transfer less on average in trust games, indicating lower levels of trust. This finding supports that of Johnson & Mislin (2011), who also found a negative relationship between being black and prosocial behaviour (trust in this case) in trust games. They proposed that South Africa’s history of Apartheid has likely negatively impacted trust levels among black South Africans.

The joint null hypothesis of all regression coefficients (other than the constants) being zero can be rejected at a 10% l.o.s. when looking at Instagram and at a 5% l.o.s. when looking at Twitter. The null hypothesis can, however, not be rejected for the combined model or Facebook.

5.3 RECIPROCITY:

Table 5 shows the coefficients and standard errors of each of the explanatory variables used in the different regressions as well as their significance level.

Table 5: Tobit Regression Results for Testing Reciprocity with 223 Observations

	Combined	Facebook	Instagram	Twitter
Reciprocity				
LikeRecent	0.040 (0.033)	0.010 (0.069)	0.037 (0.042)	0.058 (0.095)
NotLikeRecent	-0.057* (0.031)	-0.044 (0.075)	-0.030 (0.038)	-0.089 (0.091)

AlwaysLiked	0.018 (0.028)	0.133** (0.060)	-0.018 (0.036)	0.078 (0.086)
Female	-0.004 (0.026)	-0.079 (0.055)	-0.010 (0.034)	0.118 (0.099)
IncomeAbove	-0.026 (0.028)	-0.077 (0.063)	-0.006 (0.034)	-0.036 (0.142)
EthnicityB	-0.028 (0.026)	-0.008 (0.057)	-0.087** (0.037)	0.060 (0.108)
AgeOlder	0.030 (0.027)	0.137** (0.053)	0.007 (0.037)	0.092 (0.087)
Constant	0.437*** (0.027)	0.374*** (0.061)	0.468*** (0.034)	0.231* (0.116)
Observations	223	49	124	31
LR chi2	8.86	13.86	8.41	5.34
Prob > chi2	0.2628	0.0537*	0.2977	0.6181

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

NotLikeRecent was significant at a 10% l.o.s. in the combined model with a coefficient of -0.057. This shows that there is a small negative relationship between reciprocity in trust games and liking posts of others who have not liked your recent posts. This could be, because participants who would like someone's post, even if that person did not like their recent post(s), are arguably less reciprocity-based decision-makers which would, in turn, explain the weak negative relationship.

AlwaysLiked was significant at a 5% l.o.s. when looking at Facebook with a coefficient of 0.133, suggesting that there is a small positive relationship between trust game reciprocity and showing reciprocity in social media. This can be explained with a positive social history between the users. As also shown by Berg et al. (1995) and Johnson & Mislin (2011), if two participants have a positive social history with each other; there will be a greater tendency towards prosocial behaviour (reciprocating with each other).

EthnicityB was significant at a 5% l.o.s. when looking at Instagram with a coefficient of -0.087. The effect of a small negative relationship between prosocial behaviour (reciprocity in this case) and being black is seen here again. Both trust and reciprocity, therefore, have a negative relationship with being black. This again supports the findings of Johnson & Mislin (2011), with the explanation being the same as with trust; South Africa's history of Apartheid has likely negatively impacted reciprocity levels among black South Africans.

AgeOlder was significant at a 5% l.o.s when looking at Facebook with a coefficient of 0.137. This means that there is a small positive relationship between age and reciprocating between Facebook users.

The joint null hypothesis of all regression coefficients (other than the constants) being zero can only be rejected at a 10% l.o.s. when looking at Facebook. The null hypothesis can, however, not be rejected for the combined model, Instagram or Twitter.

6 DISCUSSION AND CONCLUSION

The study found that there was a clear preference for Instagram, making up 57.52% of participants, over Facebook and Twitter (21.43% and 13.53% of participants respectively).

Mean percent transfers and returns were roughly the same across all social media platforms. Mean percent transfers were slightly higher with the participants that chose Instagram and slightly lower with the participants that chose Facebook while mean percent returns were slightly higher with participants that chose Facebook and slightly lower with participants that chose Twitter. However, no differences between the mean percent transfer or returns were found to be significant even at a 10% level of significance.

Facebook users were found to be far more likely than Twitter users to like someone's post if they knew them closely (67.35% vs 56.25% of the time) and to like someone's post if that person has liked their most recent post (32.65% vs 12.5% of the time). This could indicate that Facebook users are more inclined to prosocial behaviour or that they place a greater value on social history, compared to Twitter users. Instagram users were found to be far less likely than Twitter users to comment on someone's post if they knew that person closely (4.84% vs 18.75% of the time) and far more likely than Facebook users to like someone's post if that person has not liked their most recent post (28.23% vs 12.24% of the time).

Commenting on someone's post that you don't know closely has a small negative effect on trust when looking at Facebook and a small positive effect when looking at Instagram. This could potentially be due to Instagram requesting confirmation prior to posting the post to discourage malicious comments, while Facebook does not.

Twitter users who perceive themselves as being part of an above-average income household would on average transfer more money in trust games, indicating higher levels of trust. It was further found that Black participants would transfer and return less money in trust games, indicating lower levels of prosocial behaviour.

This finding supports that of Johnson & Mislin (2011), who also found a negative relationship between being black and prosocial behaviour in trust games. This could possibly be due to South Africa's history of Apartheid which negatively impacted on trust levels amongst black South Africans.

A small negative relationship between reciprocating and someone not liking your posts was observed, therefore it indicates less of an incentive to reciprocate if your recent posts were not liked. Facebook users, on the other hand, were found to be more likely to reciprocate if someone has always liked their posts, which could potentially be as a result of a positive social history. As also shown by Berg et al. (1995) and Johnson & Mislin (2011), that two participants who have a positive social history have a greater tendency towards prosocial behaviour.

The age of participants had a small positive relationship with likelihood to reciprocate when looking at Facebook users. Older Facebook users are more likely to reciprocate than younger Facebook users which supports the findings of Sutter & Kocher (2007) as well as the findings of Van Den Bos et al. (2010) who shows that a tendency towards prosocial behaviour (trust and reciprocity) and age exist. Although it was expected that age would be significant and have a positive relationship with both trust and reciprocity, it was only found to be significant with reciprocity when looking at Facebook, probably due to the limited variation in the sample age. The fact that age and reciprocating between Facebook users were found to be significant, regardless of lack in variation of the sample age, might be indicative of a strong relationship had there been more variation in sample age.

Taking all results into account, it can be shown that there are behavioural differences between the different social media platforms and that the likelihood to trust and reciprocate differs from platform to platform. In the present age, there exists vast amounts of data and information. The popularity and usage of social media has ever been on the rise, possibly making social media behaviour an invaluable source of data. Finding a means of utilising this data to aid research could, therefore, have a monumental impact. Overall, the aspects of social media that was measured and used in this study did not have a strong enough relationship with trust game behaviour to be able to conclude that they can be used as substitutes for trust and reciprocity (as measured in the standard Berg et al. (1995) trust game). It could still, however, be possible that aspects of social media behaviour, other than the ones used in this study, could be used to predict behaviour outside of social media. A possible avenue to investigate further could be testing the impact of positive and negative comments on social media behaviour, this could have a stronger relationship with trust and reciprocity.

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8 APPENDIX

8.1 EXPERIMENT INSTRUCTIONS

You will be participating in an economic experiment called a trust game. There are 2 players in each game: a sender and a receiver. The game works as follows:

1. The sender starts with R100. He/she decides how much of this R100 to send/give to the receiver. This can be any amount (whole numbers only) from 0 to the full R100.
2. This sent amount will then be multiplied by three (by the experimenter).
3. The receiver decides how much of this tripled amount to return to the sender. Again, this can be any amount (multiples of 10 only) from 0 to the full tripled amount.

To make sure that responses are anonymous, each player has been given a random number. These numbers will be provided by the experimenter on a piece of paper. This number was randomly generated from excel. The first number will be the sending number and the second the receiving number, thus pairing groups of senders' and receivers. Since you will not know who you have been paired with until the end of the game, we ask you to play the role of both sender and receiver.

If you have any question regarding the trust game, please raise your hand. Since you are not allowed to talk during the experiment.

As the sender, you are sending part of your R100 to the person whose number is on your paper. As the receiver, you are deciding how much to return to the person who sent you some of their R100. You will not know until the end of the experiment how much that person decided to send you. Therefore, we ask you to tell us how much you would return for each possible amount that the sender might have sent you.

Please note that 5 games will be randomly selected to be paid in real money (selecting numbers from rolling a dice, after class). You should, therefore, make all decisions as if you were playing for real money. That way, if your game is one of the ones selected for payment, you will be happy with your choices. Both the sender and receiver will then be paid based on their choices in the game. If you are selected, please note you will not be paid if you have not filled in the entire trust game section and most of the other section.

8.2 TRUST GAME AND SURVEY QUESTIONS:

- **Link to study database:**

https://docs.google.com/forms/d/1FiMfiCt3HrIS6_RZkEqk1EUueTTrpFCFJNd9XMJ1vcU/preview

8.2.1 Trust Game:

1. How much of your R100 do you want to send to the player whose number is on the paper?
2. If the sender sent R10, then I received R30 and I will return R_____
3. If the sender sent R20, then I received R60 and I will return R_____
4. If the sender sent R30, then I received R90 and I will return R_____
5. If the sender sent R40, then I received R120 and I will return R_____
6. If the sender sent R50, then I received R150 and I will return R_____
7. If the sender sent R60, then I received R180 and I will return R_____
8. If the sender sent R70, then I received R210 and I will return R_____
9. If the sender sent R80, then I received R240 and I will return R_____
10. If the sender sent R90, then I received R270 and I will return R_____
11. If the sender sent R100, then I received R300 and I will return R_____

8.2.2 Facebook:

1. Will you like someone's post if you know them closely?
2. Will you like someone's post if you don't know them closely?
3. How much time do you usually spend on Facebook during a 1-day period?
4. How many friends do you have on Facebook?
5. Does the number of friends you have, matter to you?
6. Would you only post material, that would increase your number of friends?
7. If someone has liked your most recent post, will you like their recent post(s)?
8. If someone has not liked your most recent post, will you like their recent post(s)?
9. Will you like someone's recent post(s) if they have always liked your posts?
10. Will your friends like your posts if you like theirs?
11. Do you like people's posts, with the hopes that they would like yours back?
12. If someone has commented on your most recent post, will you comment on one of their posts in return?
13. If someone has not commented on your most recent post, will you comment on their recent post(s)?
14. Will you comment on someone's post(s) if you know them closely?
15. Will you comment on someone's post(s) if you don't know them closely?

16. Will you stop liking someone's posts if they haven't liked your recent post(s)?
17. If people don't ever like your posts, will you stop posting?
18. If people don't ever comment on your posts, will you stop commenting on theirs?
19. Why do you use Facebook?
20. How many times a week do you post?
21. How many times a week do you share?
22. Does the caption matter when liking or posting a post?
23. What type of material do you mostly post? (Choose the most applicable)
24. What type of posts do you most like? (Choose the most applicable)

8.2.3 Instagram:

1. Will you like someone's post(s) if you know them closely?
2. Will you like someone's post(s) if you don't know them closely?
3. How much time do you usually spend on Instagram during a 1-day period?
4. How many followers do you have on Instagram?
5. Does the number of followers you have matter to you?
6. Would you only post material, that would increase your followers?
7. If someone has liked your most recent post, will you like their recent post(s)?
8. If someone has not liked your most recent post, will you like their recent post(s)?
9. Will you like someone's recent posts if they have always liked your posts?
10. Will your friends like your posts if you like theirs?
11. If someone has commented on your most recent post, will you comment on their recent post(s)?
12. If someone has not commented on your most recent post, will you comment on their recent post(s)?
13. Will you comment on someone's post(s) if you know them closely?
14. Will you comment on someone's post(s) if you don't know them closely?
15. Will you stop liking someone's posts if they haven't liked your recent post(s)?
16. If people don't ever like your posts, will you stop posting?
17. If people don't ever comment on your posts, will you stop commenting on theirs?
18. Do you like people's posts, with the hopes that they would like yours back?
19. Do you follow celebrities or influencers?
20. Does the caption matter to you when liking or posting a photo?
21. What type of material do you mostly post? (Choose the most applicable)
22. What type of photo do you mostly like? (Choose the most applicable)
23. Do you like some people's photos without even looking at the photo?
24. In reference to question 23, who's posts are these?

8.2.4 Twitter:

- 1.** Will you like someone's tweets if you know them closely?
- 2.** Will you like someone's tweets if you don't know them closely?
- 3.** How much time do you usually spend on Twitter during a 1-day period?
- 4.** If someone has liked your most recent tweet, will you like their recent tweet(s)?
- 5.** If someone has not liked your most recent tweet, will you like their recent tweet(s)?
- 6.** Will you like someone's recent tweets if they have always liked your tweets?
- 7.** Will your friends like your tweets if you like theirs?
- 8.** If someone has commented on your most recent tweet, will you comment on their recent tweet(s)?
- 9.** If someone has not commented on your most recent tweet, will you comment on their recent tweet(s)?
- 10.** Will you comment on someone's tweet(s), if you know them closely?
- 11.** Will you comment on someone's tweet(s), if you don't know them closely?
- 12.** Will you stop liking someone's tweets if they haven't liked your recent tweets?
- 13.** If people don't ever like your tweets, will you stop tweeting?
- 14.** If people don't ever comment on your tweets, will you stop commenting on theirs?
- 15.** Will you retweet someone's tweet if you know them closely?
- 16.** Will you retweet someone's tweet if you don't know them closely?
- 17.** If someone has retweeted your most recent tweet, will you retweet their recent tweet(s)?
- 18.** If someone has not retweeted your most recent tweet, will you retweet their recent tweet(s)?
- 19.** Will you retweet someone's recent tweets if they have always retweeted your tweets?
- 20.** Will your friends retweet your tweets if you retweet theirs?